

Models used

Every model the assistant depends on: what kind of model it is, how large it is and where it runs. Two neural models are self-hosted. The browser's speech models are used only by the mimic client.

INTENT_MATCHER=llm (default)

Qwen2.5 runs both machines: machine #1 (intent matcher) and machine #2 (response formulator). Only the GPU is used.

INTENT_MATCHER=classifier

GLiNER2 and Lingua run machine #1 on the CPU, and Qwen2.5 still writes every reply as machine #2. The classifier container always starts, whichever mode is set.

Models

MODEL	TYPE OF MODEL	WEIGHT	ROLE AND WHERE IT RUNS
SELF-HOSTED, IN DOCKER COMPOSE			
Qwen2.5-7B-Instruct-AWQ Qwen team, Alibaba Cloud · Apache-2.0 huggingface.co/Qwen/Qwen2.5-7B-Instruct-AWQ	Large language model (decoder-only transformer), tuned to follow instructions. It is the 4-bit AWQ version of Qwen2.5-7B-Instruct: a third of the size, with a small loss in quality. Multilingual, including English, German, Russian and Ukrainian.	5.6 GB 7.6 B params 4-bit AWQ	GPU · vLLM Machine #2 always, and machine #1 in llm mode. It answers in schema-constrained JSON, with the context capped at 4,096 tokens. Set with LLM_MODEL.
GLiNER2.5 multi v1 Fastino · Apache-2.0 huggingface.co/fastino/gliner2.5-multi-v1	Zero-shot encoder for classification and entity extraction. It generates no text: it scores labels described in plain words and finds spans in the text, with no training. Multilingual DeBERTa-v3 encoder, 12 layers.	1.15 GB 287 M params 32-bit float	CPU · classifier Machine #1 in classifier mode. It scores the intent labels and finds form values, in under a second per turn. Set with CLASSIFIER_MODEL.
Lingua 2.1 (language detector) Peter M. Stahl · Apache-2.0 github.com/pemistahl/lingua-py	Statistical language identification built on n-gram models, not a neural network. It is designed for short texts, though a bare "ok" can still fit several languages.	≈ 96 MB 75 languages in package en, de, ru, uk loaded	CPU · classifier Provides detected_language in classifier mode. The language list is CLASSIFIER_LANGUAGES.
IN THE BROWSER, MIMIC CLIENT ONLY			
Web Speech: SpeechRecognition Google, through Chrome developer.mozilla.org · SpeechRecognition	Speech-to-text in the cloud. Chrome streams the microphone to Google's speech service. The model behind it is not published.	— cloud service needs internet	Google servers Turns the caller's voice into request_text. On a real phone line, the telephony provider's speech-to-text takes its place.
Web Speech: speechSynthesis Operating system and browser voices developer.mozilla.org · SpeechSynthesis	Text-to-speech with the voices installed for the language, such as the Microsoft voices on Windows.	— part of the OS	Local, in the browser Reads the answer aloud in the detected language.

≈ 6.8 GB

of self-hosted weights to download (one-time)

1 GPU

for vLLM: about 5.6 GB of weights, plus the KV cache

2 containers

vllm (GPU) and classifier (CPU) share one weights cache

Good to know

- Weights download on first start into LLM_CACHE_DIR, which both containers mount. Later starts are offline.
- Any model vLLM supports can replace Qwen2.5 through LLM_MODEL. Keep it instruction-tuned and multilingual, because both machines rely on it.
- dateparser, which turns "the 15th of June" into a date in classifier mode, is rule-based, not a model.